

Public Economics (ECON 131)

Section #1: Optimization Review

GSI: Steve Ramos

January 27, 2025

Contents

	Page
1 Economic Motivation	1
2 Unconstrained Optimization	1
3 Constrained Optimization	2
3.1 Mathematical tools: Substitution	3
3.2 Mathematical tools: Lagrangian optimization	3
3.3 Example 1: Homer Simpson, Pizza, and Beer	5
4 Cobb-Douglas Preferences	6
4.1 Useful properties of C-D preferences	6
4.2 Economic Implications	6
4.3 Example 1: Homer's problem, revisited	7
4.4 Example 2: Sam and labor supply	8

1 Economic Motivation

↪ we refer to this process as optimization

As you learned in Econ 100A/101A, economists generally assume that individuals seek to make choices that will maximize their utility subject to a limited set of resources (most commonly, the individual's total budget or number of hours in a day). In today's section, we will review common optimization techniques to solve problems in economics.

2 Unconstrained Optimization

↪ here: simply take the derivative of the objective function with respect to the choice variable, then set the derivative equal to zero

To solve for a function's maximum/minimum, you will need to first calculate its derivative. Most "well-behaved" functions you will encounter in economics have an interior optimum where the derivative is equal to 0.

1. Write down the function you are optimizing (called the *objective function*; e.g., the utility function).
2. Differentiate the objective function with respect to the choice variable.
3. Set the derivative equal to 0 and rearrange to solve for the choice variable as a function of everything else.

Example: Find the value of x that maximizes $f(x) = \ln(-2x) - 8x^2$.
 → our choice variable x ← our objective function $f(x)$

$$\begin{aligned}
 \textcircled{1} \quad f(x) &= \ln(-2x) - 8x^2 \\
 \textcircled{2} \quad \frac{d(f(x))}{dx} &= f'(x) = \frac{d(\ln(-2x) - 8x^2)}{dx} = \frac{d(\ln(-2x))}{dx} - \frac{d(8x^2)}{dx} \\
 &= \frac{1}{-2x} \cdot \frac{d(-2x)}{dx} - 16x = \frac{-2}{-2x} - 16x = \frac{1}{x} - 16x \\
 \textcircled{3} \quad 0 &= \frac{1}{x} - 16x \\
 &\Rightarrow 16x = \frac{1}{x} \\
 &\Rightarrow 16x^2 = 1 \\
 &\Rightarrow x^2 = \frac{1}{16} \Rightarrow x = \pm \frac{1}{4}
 \end{aligned}$$

are we done? recall: we want to maximize! which makes $f(x)$ largest?
 if $x = \frac{1}{4}$, $\ln(-2 \cdot \frac{1}{4})$ undefined, so not possible.
 thus $x = -\frac{1}{4}$

2 over what is inside of the $\ln$ times the derivative of what is inside w.r.t. x

In a more general setting, you will need to check the "second-order" conditions for an interior maximum (perhaps you have actually found a minimum?). In this class, however, we will usually restrict to a class of functions where this will always be satisfied.

In economics, we rarely start off by solving an unconstrained optimization problem. This is because when considering a normal good, the agent always prefers more to less: the unconstrained maximum for a normal good's utility function would be an infinite amount. Since individuals do not have limitless wealth, though, we are going to usually consider the scenario when individuals have to divide limited resources between at least 2 goods. This class of problems is known as *constrained optimization*.

3 Constrained Optimization

→ people do not have limitless wealth
 so we consider optimizing under certain constraints

In Econ 131 we will only deal with choices between 2 goods and will generally assume interior solutions. You will not need to worry about second order conditions. However, we will occasionally be interested in whether an individual participates in a scenario. In these cases, you should check whether choosing 0 of one good and spending all wealth on the other good has higher utility than the utility from the interior optimum that you found. We will explicitly remind you when we are working with topics where this is an issue.

The general form of the constrained optimization problems we will consider is:

budget constraint
 how much \$\$\$ they have = what they want to spend that \$\$\$ on and how much those goods cost

$$\max_{x,z} U(x,z) \quad \text{s.t.} \quad w = p_x x + p_z z$$

utility function wage price of good z price of good x budget constraint

mathematical representation of how much a person values good x and good z

} multiple ways to optimize!

objective: given their budget constraint, what bundle of good x and good z will maximize their utility

3.1 Mathematical tools: Substitution

When there are only 2 goods to choose between you can [re-write the problem as simply choosing how much of one's wealth is spent on one of the goods.] $\rightarrow$ only maximizing one good!

- Once you know how much is spent on good x , you know that the remainder of the person's budget is spent on z (i.e. $w - p_x x$ is spent on z). Thus, in a two good world, the choice boils down to how much to spend on x . $\rightarrow$ wrong! $z = \frac{w - p_x x}{p_z}$ (fix!)
- By using substitution, you can convert the problem into a single variable equation with the constraint built in, and now you can solve this the same way you solved the unconstrained maximization.

We then have:

$$w = p_x x + p_z z \Rightarrow z = \frac{w - p_x x}{p_z} \quad \downarrow \text{constraint is built in!}$$
$$\max_{x,z} U(x, z) \text{ s.t. } w = p_x x + p_z z \Leftrightarrow \max_x U\left(x, \frac{w - p_x x}{p_z}\right)$$

1. Now take the total derivative of our re-written utility function with respect to x and set it equal to 0¹
2. Solve for x to find the optimal x .
3. Substitute x into $z = \frac{w - p_x x}{p_z}$ to find the optimal z .

3.2 Mathematical tools: Lagrangian optimization

$\rightarrow$ embed the B.C. into the maximization

An alternative method for optimization is the Lagrangian method, which allows us to embed the budget constraint in the maximization. Using the same utility function and budget constraint as above, we can express the problem as follows:

$$\max_{x,z,\lambda} \mathcal{L}(x, z, \lambda) = \max_{x,z,\lambda} U(x, z) + \lambda[w - p_x x - p_z z], \quad (1)$$

where λ is a new variable which we have added to the problem, called the *Lagrange multiplier* on the constraint. A couple notes:

- We have set up $\mathcal{L}$ such that the term in brackets *should* always be weakly positive (you can not spend more than your budget). Suppose λ is positive: if you were to spend more than your budget, then your utility would go down. For this reason, we often think of λ as a "penalizer" term for violating the constraint.
- When the constraint is fulfilled and the budget is met, then the expression in brackets becomes 0 (including the multiplier). In other words, at the optimum, it must be the case that the maximum for $\mathcal{L}$ and the maximum for U coincide – so we can solve everything in terms of

¹Note that in a general setting without an explicit functional form for $U(\cdot, \cdot)$, you will need to evaluate a *total derivative*. For this class, though, you will likely get an explicit form for U .

$\mathcal{L}$! Another way to think about this is that $\mathcal{L}$ incorporates the costs of choosing more of x – namely, you can not get as much z – via inclusion of the budget constraint.

Note that $\mathcal{L}$ is now expressed in terms of an unconstrained problem, and so we can leverage the standard machinery from before. Let us take the first order conditions for maximization of $\mathcal{L}$:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial U}{\partial x} - \lambda p_x = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial U}{\partial z} - \lambda p_z = 0 \quad (3)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = w - p_x x - p_z z = 0 \quad (4)$$

The first term on the RHS of (2) is usually called the *marginal utility* in x :

$$MU_x := \frac{\partial U}{\partial x} \rightarrow \text{extra utility (or satisfaction) an individual gets from buying one more unit of good } x$$

Cancelling out λ from (2) and (3), we obtain:

$$\frac{MU_x}{p_x} = \frac{MU_z}{p_z} \quad (5)$$

Let us interpret: at an optimum, the marginal utility per dollar spent on X should be equal to the marginal utility of a dollar spent on Z . This idea is usually called the **equimarginal principle**. Let us rearrange to obtain an equivalent (and graphical) interpretation of the first order conditions:

$$MRS_{xz} := \frac{MU_x}{MU_z} = \frac{p_x}{p_z} \quad (6)$$

This expression should be familiar to you from Econ 100A/101A: the marginal rate of substitution (the rate at which a consumer is willing to trade off between two goods) must be equal to the price ratio at any *interior* optimum.²

Knowing these formulas for interior optima can be a useful shortcut for solving optimization problems!

²There are cases in which this will not work, such as: non-differentiable utility functions (e.g. perfect complements), non-strictly concave indifference curves (e.g. perfect substitutes), and negative tangency. You should be able to recognize the first two immediately, and we will need to check to make sure that we are always finding demand to be positive.

3.3 Example 1: Homer Simpson, Pizza, and Beer

Homer Simpson has \$20 to spend on pizza slices (s) and beer bottles (b). A pizza slice costs \$1, while beer costs \$2. Homer's utility is $U(s, b) = s^{1/2}b^{1/2}$. How many pizza slices and beers should Homer choose to maximize his utility? (Try solving this using substitution, the Lagrangian, and the equimarginal principle).

① Substitution (in terms of b)

what is the B.C.: $20 = 1 \cdot s + 2b$
 $s = 20 - 2b$

$$\max_b U(20-2b, b) = \max_b \underbrace{(20-2b)^{1/2}}_{f(x)} \underbrace{b^{1/2}}_{g(x)}$$

$$\frac{d[f(x)g(x)]}{dx} = f'(x)g(x) + g'(x)f(x)$$

$$\frac{d[(20-2b)^{1/2} \cdot b^{1/2}]}{db} = \frac{d[(20-2b)^{1/2}]}{db} b^{1/2} + \frac{d[b^{1/2}]}{db} (20-2b)^{1/2}$$

$$0 = b^{1/2} \cdot \frac{1}{2} (20-2b)^{-1/2} \cdot (-2) + (20-2b)^{1/2} \cdot \frac{1}{2} b^{-1/2}$$

$$\frac{b^{1/2}}{(20-2b)^{1/2}} = \frac{(20-2b)^{1/2}}{2b^{1/2}} \quad \left\{ b^{1/2} \cdot b^{1/2} = b^{1/2+1/2} = b \right\}$$

$$\Rightarrow 2b = 20 - 2b$$

$$\Rightarrow 20 = 4b$$

$$b^* = 5$$

$$s^* = 20 - 2b$$

$$= 20 - 2(5)$$

$$= 20 - 10$$

$$= 10$$

③ equimarginal principle

$$\frac{MV_s}{MV_b} = \frac{p_s}{p_b}$$

$$\frac{\frac{1}{2} \cdot \frac{b^{1/2}}{s^{1/2}}}{\frac{1}{2} \cdot \frac{s^{1/2}}{b^{1/2}}} = \frac{1}{2} \Rightarrow 2 \frac{b^{1/2}}{s^{1/2}} = \frac{s^{1/2}}{b^{1/2}}$$

$$2b = s$$

plug into B.C. s.t. $2b = 20 - 2b \Rightarrow 4b = 20$

$$s^* = 2(b) = 2(5) = 10$$

② Lagrangian Method

$$\max_{s, b, \lambda} \mathcal{L}(s, b, \lambda) = \max_{s, b, \lambda} s^{1/2} b^{1/2} + \lambda [20 - s - 2b]$$

$$\frac{\partial \mathcal{L}}{\partial s} = \frac{1}{2} s^{-1/2} b^{1/2} - \lambda = 0 \Rightarrow \lambda = \frac{1}{2} s^{-1/2} b^{1/2}$$

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{1}{2} s^{1/2} b^{-1/2} - 2\lambda = 0$$

$$\Rightarrow 2\lambda = \frac{1}{2} s^{1/2} b^{-1/2}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 20 - s - 2b = 0$$

$$2 \cdot \frac{1}{2} s^{-1/2} b^{1/2} = \frac{1}{2} s^{1/2} b^{-1/2}$$

$$\Rightarrow \frac{2b^{1/2}}{s^{1/2}} = \frac{s^{1/2}}{b^{1/2}}$$

$$\Rightarrow 2b = s$$

plug into B.C. s.t.

$$2b = 20 - 2b$$

$$4b = 20$$

$$b^* = 5$$

$$s^* = 2(b) = 2(5) = 10$$

$$MV_s = \frac{1}{2} \frac{b^{1/2}}{s^{1/2}}$$

$$MV_b = \frac{1}{2} \frac{s^{1/2}}{b^{1/2}}$$

$$2 \frac{b^{1/2}}{s^{1/2}} = \frac{s^{1/2}}{b^{1/2}}$$

$$2b = s$$

$$b^* = 5$$

4 Cobb-Douglas Preferences

Utility functions of the form $U(A, B) = A^\alpha B^\beta$ with $\alpha + \beta = 1$ are called "Cobb-Douglas" preferences. Functions of this form are used all the time in economics because they are very user friendly and have some nice properties.

- **General Form:** $U(A, B) = A^\alpha B^\beta$ with $\alpha + \beta = 1$
- **Example:** $U(A, B) = A^{0.25} B^{0.75}$

$$u(s, b) = s^{1/2} b^{1/2}$$

4.1 Useful properties of C-D preferences

1. They can be re-written as $U(A, B) = A^\alpha B^{1-\alpha}$
2. They represent two normal goods with positive diminishing marginal utility in each good.

$$\frac{\partial U}{\partial A} = \alpha A^{\alpha-1} B^{1-\alpha} \quad \frac{\partial^2 U}{\partial A^2} = \alpha(\alpha-1) A^{\alpha-2} B^{1-\alpha}$$

$$\frac{\partial U}{\partial B} = (1-\alpha) A^\alpha B^{-\alpha} \quad \frac{\partial^2 U}{\partial B^2} = -\alpha(1-\alpha) A^\alpha B^{-\alpha-1}$$

3. The optimal allocation of resources across the two goods will have a predictable form.
4. General form: Let p_A = the price of each unit of A, and p_B = the price of each unit of B. Then solving

$$\max_{A, B} U(A, B) = A^\alpha B^{1-\alpha} \quad \text{s.t.} \quad w = p_A A + p_B B$$

results in

$$s^* = \frac{1/2 \cdot (20)}{2} = 10$$

$$A^* = \frac{\alpha w}{p_A}, \quad B^* = \frac{(1-\alpha)w}{p_B}$$

$$b^* = \frac{(1-1/2)20}{2} = \frac{10}{2} = 5$$

5. One last trick: for any general utility function $U(A, B) = A^\alpha B^\beta$, the optimal A^*, B^* also solve

$$\log U(A, B) = \alpha \log A + \beta \log B$$

which is a very nice function to differentiate. This trick holds for any **monotone** (increasing or decreasing) transformation f of the original utility function.

4.2 Economic Implications

1. *This will save you time on exams!*
2. This implies that individuals with these types of preferences devote a constant share of wealth to each good. The share of wealth devoted to each good corresponds to the exponent on that good in the utility function. In the above example, the individual spends αw on A and $(1-\alpha)w$ on B.
3. Notice, the price of B does NOT appear in the formula for the optimal amount of A. Similarly the price of A appears nowhere in the optimal amount of B.

4.3 Example 1: Homer's problem, revisited

1. Knowing what we know now, how could we have solved Homer's problem in a faster way?

$$U(A, B) = A^\alpha B^\beta \text{ where } \alpha + \beta = 1, \quad U(s, b) = s^{1/2} b^{1/2} \quad 1/2 + 1/2 \text{ Cobb-Douglas!}$$

$$A^* = \frac{\alpha w}{p_A}, \quad B^* = \frac{(1-\alpha)w}{p_B}$$

$\rightarrow b^* = \frac{(1-1/2)20}{2} = \frac{10}{2} = 5$

$\downarrow$

$$s^* = \frac{1/2 \cdot (20)}{2} = 10$$

Let us try out the trick from above. Consider the alternate utility function representation:

$$\tilde{U}(s, b) = \log(U(s, b)) = \frac{1}{2} \log s + \frac{1}{2} \log b$$

2. Is this a monotonic transformation of the original utility function? Why or why not?

yes, strictly increasing

3. Find Homer's MRS using this representation. How does this relate to the MRS we found before? What does this imply for how much beer and pizza Homer chooses?

4.4 Example 2: Sam and labor supply

Sam is a substitute teacher. Every week Sam has to choose how many hours to work (L) and how many to spend on other activities (leisure). Sam earns \$10 for every hour worked and therefore has $\$10 \cdot L$ to spend on all other goods. (There are 168 hours in a week.) The price of all other goods has been normalized to 1.³ Sam's utility is: $U(c, h) = \frac{1}{6}ch - (h - 20)^2$, where h is the number of hours of leisure Sam has.

→ equimarginal principle

1. Solve for c and h using $\frac{\partial U / \partial c}{\partial U / \partial h} = \frac{P_c}{P_h}$.

what's the budget constraint?

they earn 10 for every hour (L) worked, so their income is $10 \cdot L$
they use this income to consume goods c , w/ price normalized to 1

combine s.t. BC

$$\begin{aligned} c &= 10L \\ &= 10[168 - h] \\ &= 1680 - 10h \end{aligned}$$

$$MV_c = \frac{\partial U}{\partial c} = \frac{1}{6}h$$

$$MV_h = \frac{\partial U}{\partial h} = \frac{1}{6}c - 2(h - 20)$$

$$P_c = 1$$

$$P_h = 10$$

$$\Rightarrow \frac{\frac{1}{6}h}{\frac{1}{6}c - 2(h - 20)} = \frac{1}{10}$$

$$\Rightarrow \frac{10}{6}h = \frac{1}{6}c - 2(h - 20)$$

$$\Rightarrow 10h = c - 12h + 240$$

$$\Rightarrow 22h - 240 = c$$

plug into BC

$$22h - 240 = 1680 - 10h$$

$$32h = 1920$$

$$h^* = 60$$

plug back in

$$22(60) - 240 = c$$

$$c = 1080$$

2. Now, Texas introduces a 20% wage tax. How much does Sam choose to work now?

what changes?

$$MV_c \quad \text{NO}$$

$$MV_h \quad \text{NO}$$

$$P_h \quad \text{Yes}$$

$$P_c \quad \text{NO}$$

the price of hours is the only portion impacted
instead of bringing in \$10 they bring 20% less
in, so only keep $80\% (10) = 8$

NOW

$$\frac{\frac{1}{6}h}{\frac{1}{6}c - 2(h - 20)} = \frac{1}{8}$$

$$\Rightarrow \frac{8}{6}h = \frac{1}{6}c - 2(h - 20)$$

$$\Rightarrow 8h = c - 12h + 240$$

$$\Rightarrow 20h - 240 = c$$

plug into B.C.

$$20h - 240 = 8(168 - h)$$

$$= 1344 - 8h$$

$$\Rightarrow 28h = 1584$$

$$h^* \approx 56.6$$

$$20(56.6) - 240 = c^*$$

$$c^* \approx 891$$

³Self-test: is this a restrictive assumption?